

Data Appendix

Nighttime Light Intensity as a Proxy for Economic Development

Snow Stormers | DS 4002 | Spring 2026

1.1 Dataset Overview

Each row in labels.csv represents one country's annual nighttime light composite for 2022, derived from NASA's Black Marble VNP46A4 product and stored as a normalized 64×64 float32 array in data/processed/<ISO3>_2022.npy. Of 193 World Bank-classified countries, 162 yield processable imagery; the remaining 31 are microstates that fall below the resolution threshold of the Natural Earth 110m boundary shapefile. HDF5 tiles were retrieved via the NASA CMR API (concept ID C3860065683-LAADS), mosaicked with rasterio, clipped to each country's boundary polygon, and resampled to 64×64 pixels using average resampling. Fill values, negative artifacts, and nodata pixels were set to 0, and pixel values were normalized to [0, 1] by dividing by the 99th percentile of non-zero pixels (capped at 1.0) to suppress outliers such as gas flares and offshore platforms.

Attribute	Value
Total countries (full World Bank list)	193
Countries with processable imagery	162
Countries excluded (microstates)	31
Image resolution	64 × 64 pixels (float32)
Source product	NASA Black Marble VNP46A4 (2022)
Pixel units (raw .tif)	nW/cm ² /sr
Pixel value range (normalized .npy)	[0, 1] via 99th-percentile scaling
Normalization cap	1.0 (gas flares / offshore platforms suppressed)
Spatial resolution (native)	~500 m (15 arc-second grid)
Storage format	.npy (float32) arrays + labels.csv metadata

1.2 Income Tier Distribution

Tier sizes reflect the actual World Bank 2022 country classifications rather than an artificially balanced sample. The High income tier has the most entries (62 in the full dataset); the Low tier has the fewest (27). All 27 Low-income countries have processable imagery; the 31 excluded microstates fall predominantly within the Lower-Middle and Upper-Middle tiers.

Income Tier	Countries (Full, labels.csv)	Countries (Processable, features.csv)	% of Processable
Low	27	27	16.7%
Lower-Middle	54	44	27.2%
Upper-Middle	50	44	27.2%
High	62	47	29.0%
Total	193	162	100%

1.3 Variable Descriptions — labels.csv & Image Files

Variable / File	Type	Source	Description
iso3	String	labels.csv	ISO 3166-1 alpha-3 country code. Primary key linking all datasets.
country_name	String	labels.csv	Country name from the Natural Earth 110m admin-0 boundary shapefile.
income_tier	Categorical (4 levels)	labels.csv	World Bank 2022 income classification: Low, Lower-Middle, Upper-Middle, or High. Target variable for all classification models.
<ISO3>_2022.tif	GeoTIFF float32	data/raw/	Raw clipped nighttime light raster in EPSG:4326. Pixel units: nW/cm ² /sr. nodata encoded as NaN.
<ISO3>_2022.npy	ndarray float32 (64×64)	data/processed/	Preprocessed image array. Pixel values normalized to [0, 1] via 99th-percentile scaling. Missing/nodata pixels set to 0. Used for all EDA and modeling.

2.1 Descriptive Statistics — All 162 Processable Countries

Unit of observation: Each row in features.csv represents one processable country (162 rows). It contains 12 spatial metrics computed from that country's 64×64 normalized nighttime light array, plus the three identifying fields from labels.csv (iso3, country_name, income_tier). These features serve as inputs to the logistic regression baseline classifier and informed the exploratory data analysis. Spearman rank correlations (ρ) were computed between each metric and ordinal income tier rank (Low=0, Lower-Middle=1, Upper-Middle=2, High=3). Significance levels are denoted: *** $p < 0.05$.

Metric	Mean	Std Dev	Min	25th Pct	Median	75th Pct	Max
mean_brightness	0.0179	0.0204	0.0000	0.0051	0.0125	0.0235	0.1851
total_light	73.47	83.49	0.00	20.99	51.40	96.21	758.08
p90_brightness	0.0341	0.0618	0.0000	0.0002	0.0082	0.0471	0.5766
lit_fraction	0.1415	0.1371	0.0000	0.0369	0.0917	0.2051	0.6204
light_density	0.1311	0.0373	0.0000	0.1108	0.1242	0.1393	0.3632
spatial_dispersion	0.2998	0.0562	0.0000	0.2703	0.3064	0.3333	0.4214
hotspot_count	38.70	30.55	0.00	17.00	32.00	50.75	180.00
gini	0.7898	0.1022	0.0000	0.7523	0.8099	0.8542	0.9400
entropy	3.4795	0.4512	0.0000	3.2745	3.4806	3.7125	5.2184
urban_concentration	0.3566	0.0625	0.0000	0.3246	0.3615	0.3914	0.5618

2.2 Per-Tier Summary Statistics — mean $\pm$ std

Income Tier (n)	Mean Brightness	Lit Fraction	Gini	Hotspot Count	Entropy
Low (n=27)	0.0042 $\pm$ 0.0031	0.030 $\pm$ 0.023	0.839 $\pm$ 0.043	14.5 $\pm$ 9.8	3.313 $\pm$ 0.380
Lower-Middle (n=44)	0.0129 $\pm$ 0.0140	0.104 $\pm$ 0.117	0.788 $\pm$ 0.146	30.2 $\pm$ 22.1	3.404 $\pm$ 0.631
Upper-Middle (n=44)	0.0154 $\pm$ 0.0099	0.131 $\pm$ 0.093	0.808 $\pm$ 0.061	41.4 $\pm$ 25.1	3.472 $\pm$ 0.241
High (n=47)	0.0330 $\pm$ 0.0283	0.251 $\pm$ 0.154	0.747 $\pm$ 0.092	58.0 $\pm$ 37.0	3.653 $\pm$ 0.390

- Figure 1: Pixel Intensity Distribution by Income Tier**

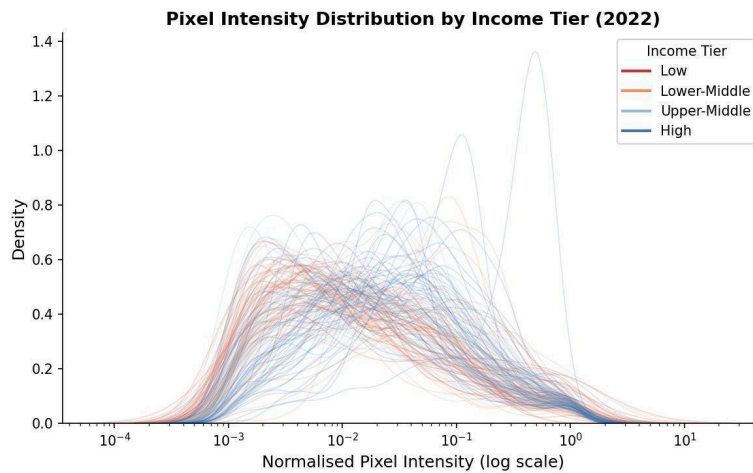


Figure 1. KDE curves on a log-scaled x-axis showing per-country pixel intensity distributions by income tier. Low-income countries (red) cluster near 10^{-3} with limited rightward spread. High-income countries (dark blue) show heavier tails extending to 10^{-1} and beyond. Substantial overlap between Lower-Middle and Upper-Middle tiers is visible throughout.

- Figure 2: Average Nighttime Light Brightness by Income Tier**

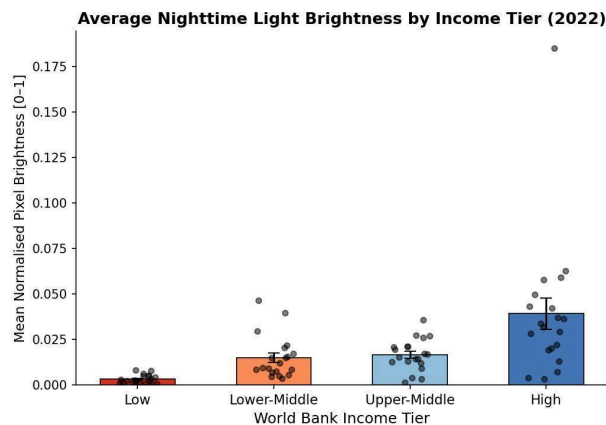


Figure 2. Bar chart with overlaid country-level dots confirming a monotonic increase in mean normalized brightness from Low to High. The High tier shows the greatest within-tier variance, including one outlier near 0.185. Lower-Middle and Upper-Middle distributions overlap substantially, consistent with the weak metric separation observed in EDA.

- **Figure 3: Sample Nighttime Light Images by Income Tier**

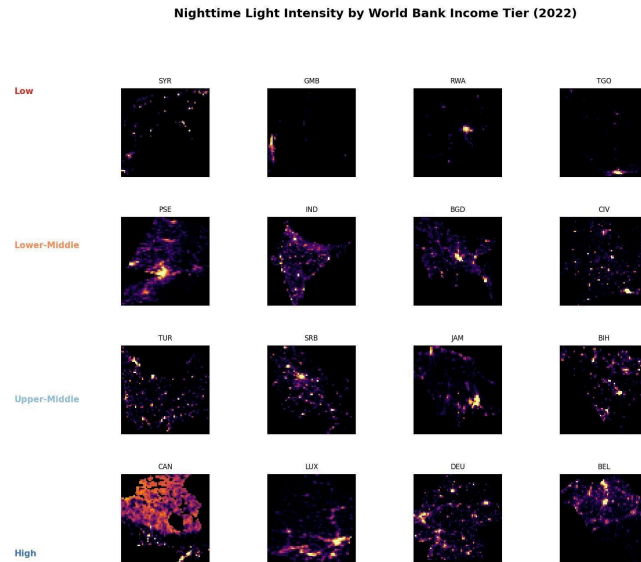


Figure 3. Representative 64×64 normalized images for four countries per tier (sorted by mean brightness within tier). Low-income countries (SYR, GMB, RWA, TGO) are nearly entirely dark with sparse point sources. Lower-Middle countries (PSE, IND, BGD, CIV) show scattered bright nodes. Upper-Middle countries (TUR, SRB, JAM, BIH) display more distributed illumination. High-income countries (CAN, LUX, DEU, BEL) show broader, more continuous light coverage.

- **Figure 4: Spearman Correlation Heatmap**

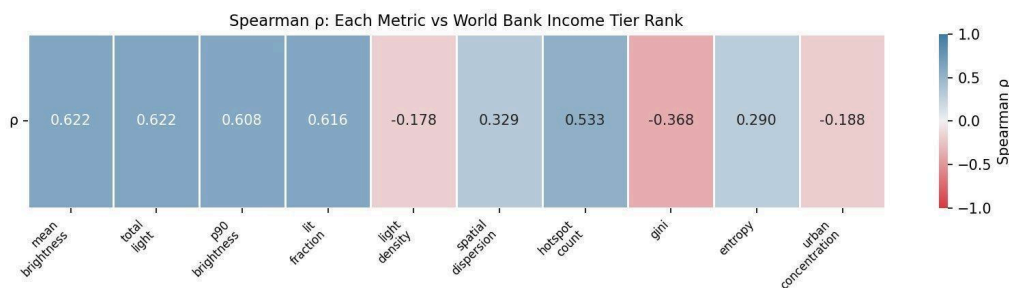


Figure 4. Spearman ρ values for all derived metrics against ordinal tier rank (Low=0 to High=3). Mean brightness, total light, lit fraction, and hotspot count are the strongest positive predictors. Gini is the strongest negative predictor. Light density and urban concentration are weakly negative, reflecting higher intensity-per-lit-pixel in sparser, lower-income countries.

2.3 Variable Descriptions — features.csv

The table below documents all 12 derived spatial metrics stored in features.csv, organized by category: Intensity metrics measure how bright the country is; Coverage metrics measure how much of the territory is illuminated; Spatial metrics characterize the geographic distribution of light; and Inequality metrics measure how evenly light is distributed across pixels.

Variable	Category	Description	Spearman ρ
mean_brightness	Intensity	Mean normalized pixel value across the full 64×64 array. Primary brightness indicator.	0.622***
total_light	Intensity	Sum of all pixel values; proxy for total radiance output. Correlated with country area.	0.622***
p90_brightness	Intensity	90th-percentile pixel value; floor for the brighter portions of the country.	0.608***
lit_fraction	Coverage	Proportion of pixels exceeding the 0.01 lit threshold. Measures share of territory with detectable light.	0.616***
light_density	Coverage	Mean pixel value restricted to lit pixels (>0.01). Intensity per lit unit of area. Weakly negative: lower-income countries concentrate light in fewer, brighter spots.	−0.178*
spatial_dispersion	Spatial	Standard deviation of lit-pixel positions, normalized to [0,1]. Higher = light is spread across more of the country.	0.329***
center_of_mass_x	Spatial	Light-weighted horizontal center of mass (0=left edge, 1=right edge). No significant tier signal. Excluded from modeling.	—
center_of_mass_y	Spatial	Light-weighted vertical center of mass (0=top, 1=bottom). No significant tier signal. Excluded from modeling.	—
hotspot_count	Spatial	Number of distinct bright clusters (connected components with mean pixel ≥ 0.1). Larger, more economically active countries have more lit urban areas.	0.533***
gini	Inequality	Gini coefficient of full pixel intensity distribution. High = light concentrated in few bright pixels. Decreases with tier: higher-income countries illuminate territory more uniformly.	−0.368***
entropy	Inequality	Shannon entropy of pixel intensities (bits). Higher = more diverse intensity values and more complex light landscape.	0.290***
urban_concentration	Inequality	Share of total light emitted by the top-5% brightest pixels. Weakly negative: dominant in lower-income countries with primate-city patterns.	−0.188*

Figure 5: Logistic Regression Coefficients

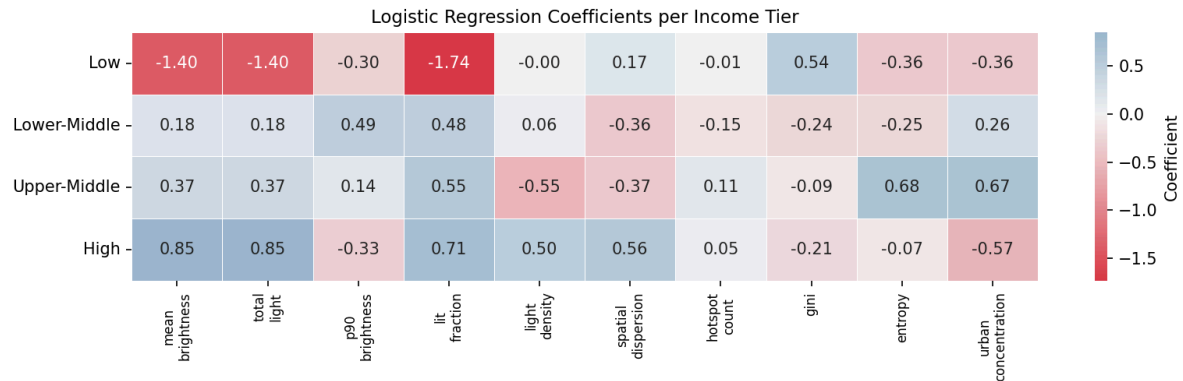


Figure 5: The coefficient heatmap reveals that mean brightness, total light, and lit fraction are the strongest positive predictors of the High tier and the strongest negative predictors of Low, aligning with EDA Spearman correlations. The Gini coefficient loads positively for Low (concentrated light) and negatively for High (uniform light), while entropy and urban concentration show opposing patterns between Upper-Middle and High tiers, suggesting these inequality metrics help separate the upper two classes.

Figure 6: Confusion Matrices

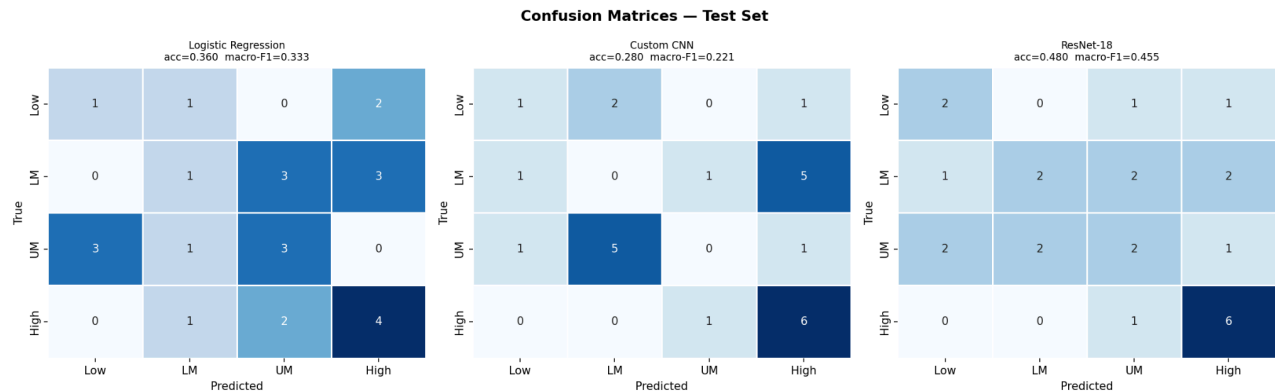


Figure 6: ResNet-18 achieves the best performance (acc=0.480, macro-F1=0.455), followed by Logistic Regression (acc=0.360, macro-F1=0.333), with the Custom CNN performing worst (acc=0.280, macro-F1=0.221). Across all three models, the High income tier is the most consistently classified correctly, while the Lower-Middle and Upper-Middle tiers show the most confusion with each other — a pattern consistent with their overlapping light distributions identified in EDA.